

RYAN TAN

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Education

B.S. in Computer Science in Immersive Media and Game Development

DigiPen Institute of Technology

Aug 2022 - Apr 2026

Diploma in Information Technology

Ngee Ann Polytechnic

Apr 2017 - Mar 2020

Work Experience

Front-End Developer - Dynamite Games

TypeScript | HTTP | Cocos Creator | Front-end | Localization

May 2025 - Apr 2026

- Built prototypes and developed a new product within development team.
- Implemented localisation and player-facing features across regional releases.
- Worked on production codebases and HTTP services to support product functionality.
- Optimised memory usage and researched new technologies to improve existing products.

Robotic Process Automation Developer - PSA Singapore

C# | Automation | UiPath | Documentation

Mar 2019 - Aug 2019

- Developed and maintained business process automation solutions using UiPath.
- Enhanced and supported existing automated workflows to improve operational efficiency.
- Produced business process documentation to support deployment and maintenance.

Projects

Real-Time Strategy Pathfinding (Unreal Engine) - Massive

C++ | Unreal Engine 5 | Algorithms | AI | Pathfinding

May 2025 - Apr 2026

- Designed and implemented graph search and pathfinding algorithms for large-scale autonomous unit navigation including A*, Lazy Theta*, Flow Fields and Boids local avoidance.
- Simulated, recorded and evaluated large-scale unit movement to compare algorithm performance.

Custom C++ Game Engine - Snookerina

C++ | C# | Custom Engine | Mono

Sep 2024 - Apr 2025

- Developed engine systems within a custom C++ game engine in a 13-person multidisciplinary team.
- Integrated Mono C# scripting runtime to support gameplay scripting for development iteration.

Custom C++ Game Engine - BoneFire Mayhem

C++ | C# | Custom Engine | Mono | Wwise

Aug 2023 - Mar 2024

- Developed engine systems within a custom C++ game engine in a 8-person multidisciplinary team.
- Integrated Mono C# scripting runtime to support gameplay scripting for development iteration.
- Integrated Wwise audio middleware and designed re-usable math, physics systems.

Capstone Project (Unity Engine) - Obliq

C# | Unity Engine | Game Design

Oct 2019 - Mar 2020

- Developed gameplay systems, UI, VFX and audio using Unity.
- Designed gameplay mechanics and level interactions with Unity's scripting and particle system.

Global Game Jam 2026 (Unity Engine) - Mask the Damage

C# | Unity Engine | Game Design

Feb 2026

- Collaborated within 5-person team to design, develop and deliver game prototype within 72-hours.
- Implemented gameplay systems and core mechanics under rapid development.

Skills

Programming Languages C++, C#, C, Python, TypeScript, HTTP

Frameworks Unreal Engine, Unity, Cocos Creator

Technologies Mono Embedding, Wwise

Tools JIRA, Figma, GitHub, Visual Studio, Rider, Linux (WSL), Valgrind, CMake